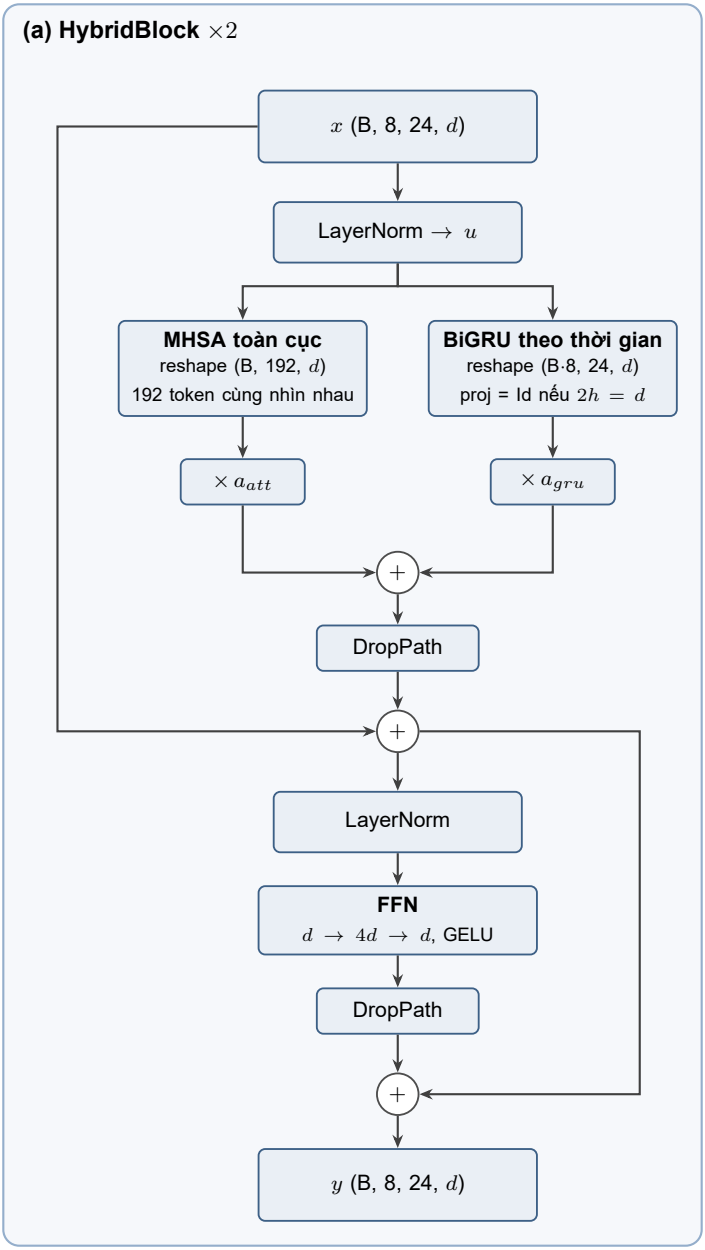
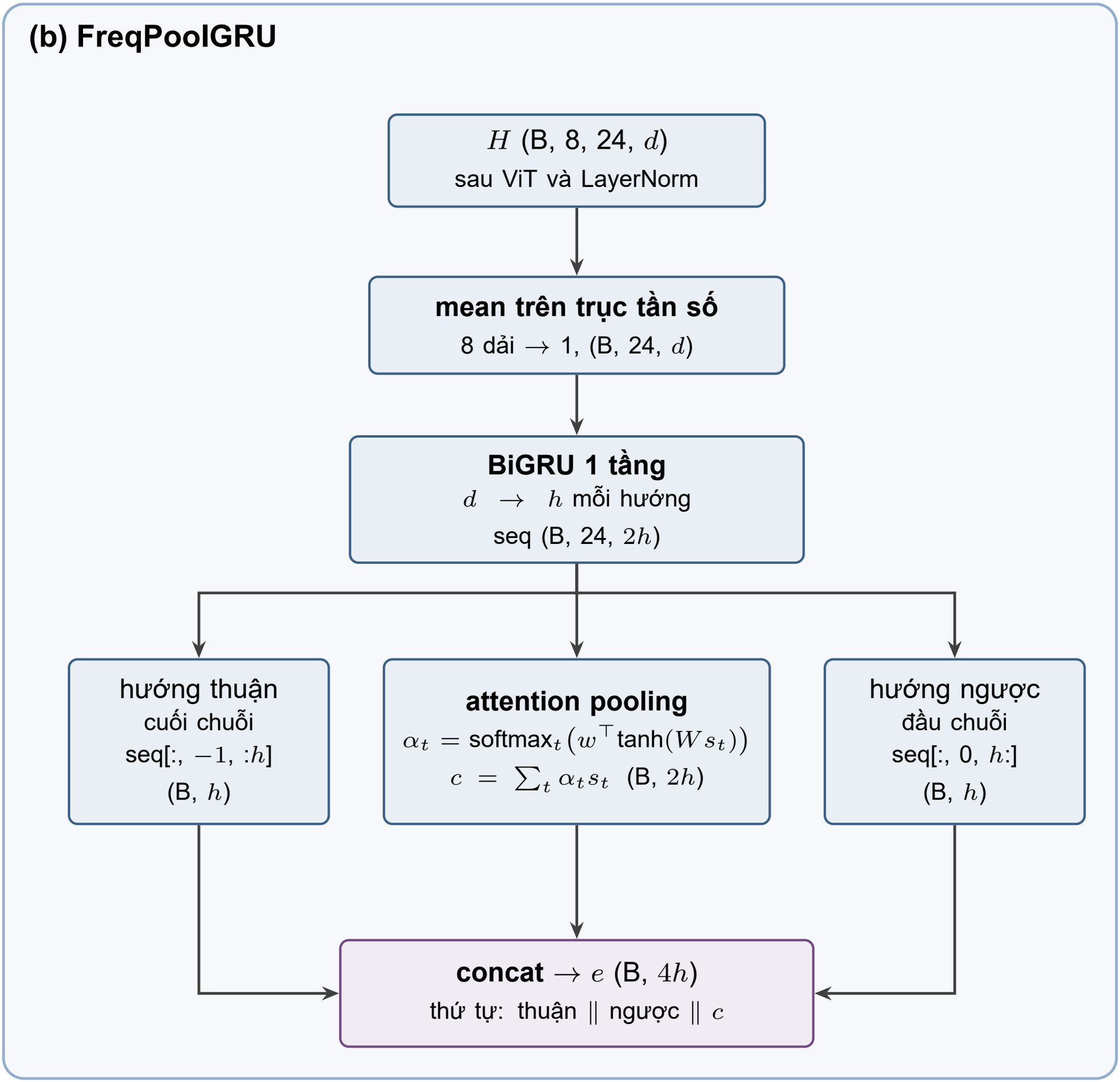
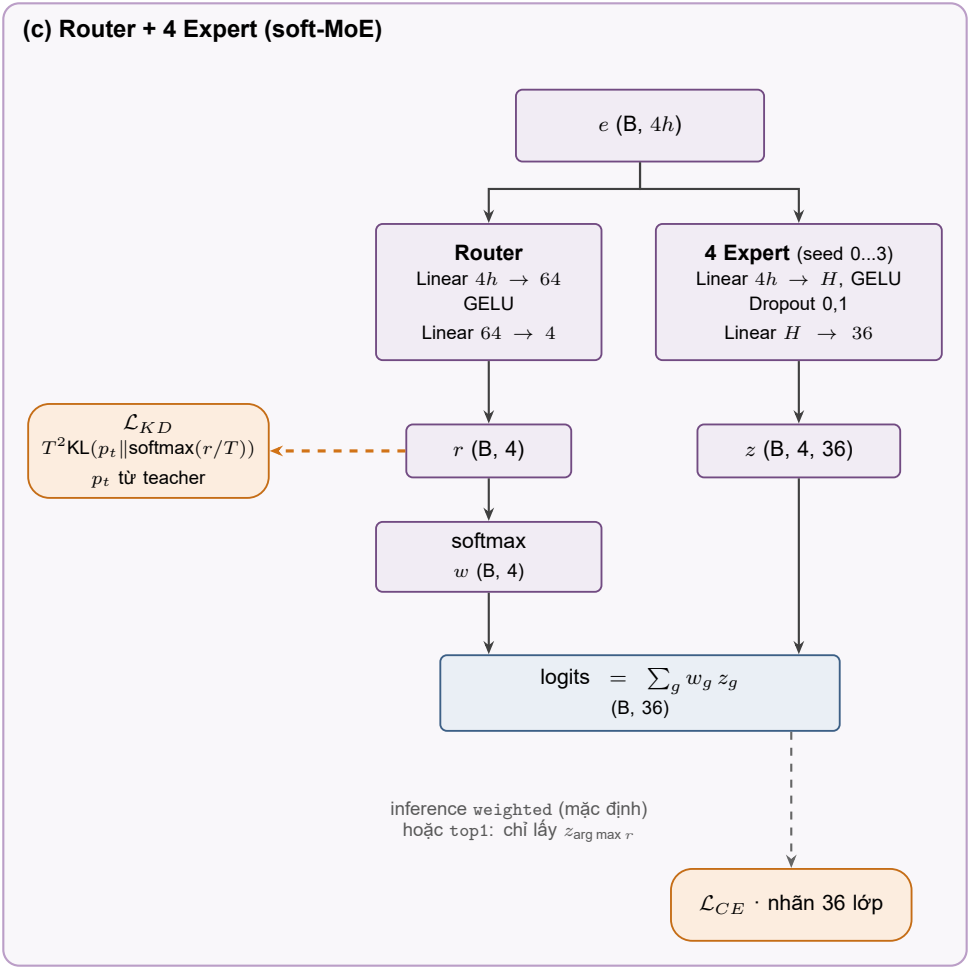

(a) HybridBlock $\times 2$
x (B, 8, 24, d)
LayerNorm $\rightarrow$ u
MHSA toàn cục
reshape (B, 192, d)
192 token cùng nhìn nhau
BiGRU theo thời gian
reshape (B·8, 24, d)
proj = Id nếu $2h = d$
$\times a_{att}$
$\times a_{gru}$
+
DropPath
+
LayerNorm
FFN
$d \rightarrow 4d \rightarrow d$, GELU
DropPath
+
y (B, 8, 24, d)


(b) FreqPoolGRU
H (B, 8, 24, d)
sau ViT và LayerNorm
mean trên trục tần số
8 dải $\rightarrow$ 1, (B, 24, d)
BiGRU 1 tầng
$d \rightarrow h$ mỗi hướng
seq (B, 24, $2h$)
hướng thuận
cuối chuỗi
seq[:, -1, :h]
(B, h)
attention pooling
$\alpha_t = \text{softmax}_t\left(w^\top \tanh(W s_t)\right)$
$c = \sum_t \alpha_t s_t$ (B, $2h$)
hướng ngược
đầu chuỗi
seq[:, 0, h:]
(B, h)
concat $\rightarrow e$ (B, $4h$)
thứ tự: thuận || ngược || c


(c) Router + 4 Expert (soft-MoE)
e (B, 4h)
Router
Linear 4h → 64
GELU
Linear 64 → 4
4 Expert (seed 0...3)
Linear 4h → H, GELU
Dropout 0,1
Linear H → 36
L_KD
T² KL(p_t ‖ softmax(r/T))
p_t từ teacher
r (B, 4)
z (B, 4, 36)
softmax
w (B, 4)
logits = Σ_g w_g z_g
(B, 36)
inference weighted (mặc định)
hoặc top1: chỉ lấy z_arg max r
L_CE · nhãn 36 lớp